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Exploring AI potential in optical analysis of organic and elemental carbons: A review of emerging applications in airborne particulate characterization

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ABSTRACT

Carbonaceous aerosols, comprising organic carbon (OC) and elemental carbon (EC), constitute up to 70% of atmospheric particulate matter (PM) and significantly impact human health, climate forcing, and air quality. Despite their significance, accurate characterization of OC and EC fractions remains challenging due to aerosols' complex and variable composition and the constraints in conventional analytical techniques. This review aims to assess how artificial intelligence (AI) can advance OC/EC characterization by enhancing data interpretation, automation, and measurement reliability. Recent developments in photo-optical methods for OC/EC quantification, including real-time monitoring, destructive, and non-destructive techniques, are comprehensively analyzed. The review further examines emerging efforts to integrate AI into conventional analytical frameworks, critically evaluating the potential of semi-supervised learning, active learning, incremental learning, TinyML, explainable AI, and large language models (LLMs) for improving OC/EC analysis. Notably, the underexplored potential of these AI approaches in source apportionment and real-time monitoring is emphasized, alongside a proposed roadmap for implementing AI-driven strategies in carbonaceous aerosol analysis and air quality assessments. AI integration with existing OC/EC analytical techniques can minimize measurement uncertainties and enable real-time, cost-effective monitoring systems. This approach facilitates high-resolution source apportionment with broad applications in environmental monitoring research, ultimately improving urban air quality diagnostics and evidence-based pollution control strategies.

1. Introduction

Artificial intelligence (AI) has emerged as a transformative technology in modern scientific research, revolutionizing data analysis, predictive modeling, and complex problem-solving across multiple disciplines (Rashid and Kausik, 2024). This paradigm shift includes various algorithmic approaches, such as deep learning neural networks, ensemble methods, and reinforcement learning techniques, which enhance the extraction of significant patterns from multivariate datasets. Integrating AI methodologies has streamlined traditionally time-intensive analyses into optimized computational processes, enabling hypothesis generation and validation (Mo et al., 2019). The rapid evaluation of AI-driven methodologies has enabled unprecedented processing of large-scale datasets and unlocked new frontiers in environmental research. Modern environmental monitoring systems generate terabytes of diverse data daily, including satellite imagery, ground-based sensor networks, and mobile monitoring platforms. This creates analytical challenges that exceed the capabilities of traditional

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statistical frameworks. AI systems excel at identifying spatiotemporal correlations within these multidimensional datasets, revealing previously undetectable patterns in atmospheric composition, pollutant transport dynamics, and emission source characteristics (Balamurugan et al., 2023).

Conventional approaches, such as statistical regressions and chemical transport models, are increasingly constrained by several critical limitations. For instance, data gaps can arise from sparse monitoring networks or incomplete observational records. Additionally, substantial computational resource requirements make them impractical for real-time or high-resolution applications and hinder their adaptability to the atmospheric processes' dynamic and nonlinear nature, reducing their predictive accuracy under varying environmental conditions.

Advancements in big data analytics and computational power have positioned machine learning (ML) and AI as resource-efficient alternatives for multiple applications, including source apportionment, particulate matter (PM) classification, and emission trend forecasting. These methods automate feature extraction, identify complex patterns, and minimize reliance on manual parameterization. The implementation of AI frameworks has shown transformative potential in atmospheric science research, including aerosol properties retrieval through remote sensing and lidar technologies, enhanced source apportionment accuracy, and high-resolution real-time air quality forecasting (Jayaraman et al., 2025; Redemann and Gao, 2024; Xu et al., 2023), as well as modeling wildfire spread (Hung et al., 2025) and improving air quality management strategies (Kaginalkar et al., 2021).

ML applications in air quality research mainly focus on forecasting and predicting air pollutant concentrations, such as PM, nitrogen and sulfur oxides (SO_2 and NOx), and ozone (O_3) (Chen et al., 2018; Feng et al., 2015; Hu et al., 2017; Li et al., 2017). Data-driven models offer several advantages over physical models, leveraging large real-world datasets to identify complex relationships between input parameters (e.g., meteorological data, emissions inventories, and land use characteristics) and output variables (e.g., air pollutant concentrations) while requiring fewer computational resources and exhibiting higher adaptability (Berkani et al., 2023; Ma et al., 2020). However, the performance of these models is critically dependent on the training data quality, comprehensiveness, and representativeness, as well as the relevance of the selected features. For example, studies have shown that incorporating supplementary parameters, such as the co-pollutant concentrations and temporal variations in land use, significantly improves the accuracy of ML models for forecasting $PM_{2.5}$ concentrations (Peng et al., 2024; M. Wang et al., 2022; Wong, 2015; Zheng et al., 2016).

Integrating physical models with ML methods enhances air quality research (Hickman et al., 2025) by identifying patterns in extensive atmospheric and environmental datasets related to pollutants (Cao et al., 2025) and can also emulate components of atmospheric models, alleviate computational bottlenecks, and create simplified, reduced-order models (See and Adie, 2021).

Physics-Informed Neural Networks (PINNs) are an emerging ML approach that embeds physical laws directly into the model's architecture, balancing computational speed with accuracy and addressing error instabilities often observed in traditional ML emulation (Chen et al., 2025). PINNs can potentially reduce computational burdens (Hu et al., 2024) and learn from diverse datasets, including both observational and numerical modeling data (Abdallah et al., 2014).

Beyond forecasting applications, AI techniques have also been applied to source apportionment, identifying emission sources and quantifying their respective contributions to air pollutant concentrations (Hopke et al., 2020). Traditional statistical methods, including positive matrix factorization (PMF) and UNMIX, are standard tools but exhibit notable limitations, requiring prior knowledge of the source numbers, relying heavily on expert judgment for source profile identification, and do not capture non-linear source-concentration relationships (Cai et al., 2022; Henry and Christensen, 2010; Hopke, 2016; Srivastava et al., 2021).

ML methods have been effectively integrated with receptor models (PMF and UNMIX) to enhance performance and overcome limitations (Ahlberg et al., 2017; Rutherford et al., 2021). PMF outputs serve as inputs for ML models, enabling prediction of source contributions from high-frequency or real-time data (Xu et al., 2024). Explainable ML tools such as SHAP (SHapley Additive exPlanations) and PDP (Partial Dependence Plots) further reveal relationships between pollutants, meteorology, and source impacts. For example, Xiao et al. used PMF-XGBoost with SHAP to link volatile organic compounds (VOCs) to ozone formation, while Z. Zhang et al. (2022) applied Random Forest (RF) with SHAP to assess $PM_{2.5}$ sources in Tianjin, China (Xiao et al., 2025). Moreover, sequential ML approaches can also help distinguish local from transported emissions (Lei et al., 2025) using rolling PMF and support vector regression (SVR). Another application of sequential approaches uses ML models to reduce subjectivity in the PMF source identification step. For example, a k-nearest neighbors (KNN) model, trained on established source profiles, has been applied to label factors derived from factor analysis (FA) models, resulting in more consistent and reproducible source identifications (Kumar et al., 2023). AI applications in air quality research extend to environmental epidemiology to predict health outcomes associated with air pollution exposure (Araujo et al., 2020; Cazzolla Gatti et al., 2020; Hill et al., 2019; Polezer et al., 2018; M. Wang et al., 2022).

Despite significant advances in computational capabilities, identifying and characterizing atmospheric particles' chemical compositions remains a significant challenge in air pollution research, requiring extensive resources and labor-intensive methodologies. A critical research gap persists in the application of AI and ML for identifying organic carbon (OC) and elemental carbon (EC) in airborne PM, and a distinct lack of comprehensive reviews exploring AI's potential in this domain.

To the best of our knowledge, no studies specifically examine the application of AI/ML to the characterization of optical parameters of OC/EC. Existing reviews in the air pollution domain address diverse topics, including AI and data mining approaches in air pollution epidemiology (Bellinger et al., 2017) and links between air pollution and health outcomes (e.g., chronic airway diseases). They also examine climate and heat-wave prediction (Subramaniam et al., 2022); deep learning models for forecasting pollutant concentrations (Ayturan et al., 2018), and supervised learning for key air pollutant prediction (Essamlali et al., 2024). Further focus areas include indoor air quality prediction with emphasis on CO_2, $PM_{2.5}$, PM_{10}, VOCs, and formaldehyde, with sustainability-oriented implications (Latoń et al., 2025); comparative evaluations of AI methods for forecasting air quality episodes (Masood and Ahmad, 2021); integration of AI with geospatial data and data mining to assess combined air pollution and health impacts (Chadalavada et al., 2025); and remote sensing coupled with ML/DL for air quality monitoring (Bahadur et al., 2023). Although most existing reviews provide comprehensive assessments of recent AI applications, they largely emphasize concentration prediction and modeling of major pollutants. Moreover, they typically focus on $PM_{2.5}$, PM_{10}, CO, CO_2, NO_2, SO_2, and O_3 as input parameters. In contrast, this review centers on carbonaceous aerosols characterized by complex chemistry which demands dedicated analysis and thorough investigation of current AI trends in their characterization, highlighting methodological advances and remaining challenges.

Existing reviews also mainly have focused on the methodological and interpretive foundations of carbonaceous aerosol analysis, summarizing established optical methods for OC/EC characterization, including thermal-optical analysis (TOA), filter-based light absorption, in situ optical techniques, laser-induced incandescence (LII), and related approaches (Correa-Ochoa et al., 2023; Duarte et al., 2022; Kong et al., 2024; Massabò and Prati, 2021; Wu et al., 2018, 2025; Zhang et al., 2023).

More recent work has introduced absorption-enhancement modeling, AI-assisted OC/EC classification, and an indirect analytical approach (Kong et al., 2024; Wu et al., 2025). This review provides a comprehensive analysis of destructive and non-destructive techniques for OC/EC quantification, integrating evidence on successful AI/ML applications that enhance measurement accuracy. Beyond summarizing current practice, it introduces AI/ML strategies that have not yet been applied to OC/EC but hold substantial practical potential. The paper is intended to serve as both a reference and a roadmap for future research and translation to practice.

Carbonaceous aerosols, classified as OC, EC, or black carbon (BC), and inorganic carbon, constitute a significant fraction (20–70%) of atmospheric PM and have long been a subject of environmental and public health research. These components significantly influence global warming, visibility degradation (Omarova et al., 2025), and human health (Aswini et al., 2018; Correa-Ochoa et al., 2023; Le et al., 2020; Massabò and Prati, 2021; Zotter et al., 2017), with EC/BC demonstrating strong epidemiological associations with respiratory and cardiovascular hospitalizations (Basagaña et al., 2015; Ostro et al., 2015; Yang et al., 2023).

AI offers the potential to address fundamental limitations in optical methods used for OC/EC characterization. Current optical analysis techniques, including thermal-optical analysis (TOA), aethalometers, and multi-wavelength absorption analyzers, require extensive calibration to mitigate filter-loading biases, light scattering, and matrix effects. AI and ML-driven models can optimize these corrections and improve spectral deconvolution capabilities, enhancing the discrimination of OC/EC fractions (Li and May 2022). In optical-scattering-based aerosol properties measurement methods, current retrieval algorithms often face accuracy or speed limitations that become increasingly restrictive as instrumentation technology advances and measurement complexity grows. However, ML frameworks provide new opportunities to overcome these challenges by simultaneously achieving computational speed and analytical precision (Boiger et al., 2022). For TOA, Shen et al. (2025) proposed AI-driven interpretation approaches that provide novel perspectives on thermogram data interpretation and carbon fraction discrimination. Similarly, spectrum analysis benefits substantially from AI implementation through automated peak identification and pattern recognition capabilities.

These applications facilitate improved identification (Khuzestani et al., 2025) and prediction of OC/EC fractions, overcoming the limitations of manual spectrum interpretation. ML models enhance aerosol species detection and retrieval of parameters such as optical depth and layer height from satellite data, improving aerosol characterization (Natraj et al., 2022; Vasilatou et al., 2025).

This research addresses critical knowledge gaps by first conducting a literature review through three systematic components: 1) a comprehensive analysis of photo-optical methodologies with high potential for AI integration; 2) critical evaluation of current AI applications in atmospheric particle characterization; and 3) development of an implementation roadmap to enhance OC and EC identification protocols, ultimately advancing air pollution assessment accuracy. This literature review evaluates key carbon-containing PM constituents and their associated optical properties, provides a comprehensive review of existing real-time monitoring and measuring methods, filter-based destructive techniques, and non-destructive optical methods for OC/EC characterization. It also discusses the current AI applications relevant to the outlined analytical techniques and assesses the feasibility of AI-driven methodologies for $PM_{2.5}$ carbon characterization, featuring practical implementations through case studies and proposing optimized frameworks for data integration and pattern recognition.

2. Optical metrics for characterizing carbonaceous particles

The carbonaceous fraction of PM consists of a complex and diverse mixture of organic compounds, presenting considerable analytical challenges for accurate constituent quantification (Wu et al., 2018). OC in the atmosphere comprises both primary OC (POC) and secondary OC (SOC) (Fig. 1). POC is directly emitted from fossil fuel combustion and biomass burning sources, while SOC forms through the atmospheric oxidation of VOCs (Jimenez et al., 2009; Karanasiou et al., 2015). Overall, OC predominantly exhibits scattering properties due to the prevalence of non-light-absorbing components, resulting in a net cooling effect on Earth's radiative balance. However, a fraction of OC known as brown carbon (BrC) has significant light-absorbing properties. BrC contributes to positive radiative forcing and modifies climate dynamics (Katoch et al., 2024).

EC, often used interchangeably with BC or soot in environmental studies, is a primary pollutant emitted from the incomplete combustion of carbon-based fuels and is recognized as the second-largest contributor to global warming after CO_2 (Bai et al., 2023; Dumka et al., 2018; Stocker, 2013; Surendran et al., 2013). The distinction between EC and BC depends on the quantification method: EC is a carbon-only substance measured by thermal or thermal-optical methods, whereas BC refers to the light-absorbing carbon fraction quantified by optical techniques (Liu et al., 2022). BC is often reported as equivalent BC (eBC) when derived from optical attenuation measurements using a fixed mass absorption cross-section (Chow et al., 2004). Alternatively, refractory BC (rBC) represents the carbon mass measured from the thermal emission of carbon mass absorbing laser energy (Lack et al., 2014). The strong light-absorbing properties of EC/BC contribute to reduced visibility, altered precipitation patterns, and decreased agricultural productivity (Bai et al., 2023; Ismail et al., 2021; Jing et al., 2019; Zotter et al., 2017).

The light-absorbing behavior of carbonaceous particles is assessed using a range of instruments, including aethalometers, spectrophotometers, thermo-optical, and absorbance-based analyzers, which yield key optical metrics (e.g., light absorption coefficient (b_{abs}), mass absorption cross-section (MAC), Ångström exponents, absorption enhancement factor (E_{abs})) (Kong et al., 2024).

The absorption coefficient (b_{abs} or σ_{abs}) (Mm^{-1}), known as the volumetric absorption coefficient, describes the absorbance of a sample per unit optical path length. The reduction in light intensity as it passes through the sample is determined by the ratio of incident to transmitted light, following the Beer-Lambert law (Kong et al., 2024). At 370 nm, BC generally exhibits stronger absorption than BrC. For example, Bai et al. (2023) report 13.72 ± 10.59 Mm^{-1} for BC vs. 3.3 ± 3.3 Mm^{-1} for BrC, and Tao et al. (2024) report 70.8 ± 47.6 Mm^{-1} vs. 53.6 ± 45.7 Mm^{-1} for BC and BrC, respectively.

The differences in absorption characteristics are quantified by Mass absorption efficiency (MAE) (m^2g^{-1}), estimates of which are sensitive to pollution sources, chemical constituents, aerosol mixing states, and meteorological conditions (Samiksha et al., 2022). For consistency, MAE is calculated by dividing the b_{abs} by the concurrently measured mass concentration of BC, ensuring identical conditions for obtaining both measurements. For BC, Wu et al. (2024) reported a maximum MAE of m^2/g at 550 nm, noting that organic coating on BC particles enhances absorption. Katoch et al. (2024) identify BC as the only absorbing species at 880 nm, with a peak MAE of 18.47 m^2 g^{-1} at 370 nm. Moreover, C. Li et al., 2024 report the highest MAE values for BC of 10.4 ± 0.4 m^2 g^{-1} at 520 nm and 10.3 ± 0.2 m^2 g^{-1} at 880 nm, with BC remaining the sole absorber at 880 and 950 nm.

Ångström exponents, including extinction Ångström exponent (EAE), absorption Ångström exponent (AAE), and single-scattering albedo Ångström exponent (SSAAE) represent the relationship between particle size, composition, and particle absorption behavior (Kong et al., 2024; Zhang et al., 2024). The association between the optical properties and size distribution is reflected in the EAE, where EAE >1 indicates fine-mode dominance, and EAE <1 suggests coarse-mode particles (Ramachandran and Rupakheti, 2022; Tiwari et al., 2015). The AAE describes the absorption's wavelength (λ (nm)) dependence and is used to differentiate absorbing aerosol types. This method assumes that BC absorbs strongly in the near infrared (e.g., 880 nm) with an AAE ≈ 1.0, allowing BrC absorption to be inferred at shorter wavelengths (Fig. 2). BC mixed with non-absorbing materials may exhibit an AAE up to 1.6

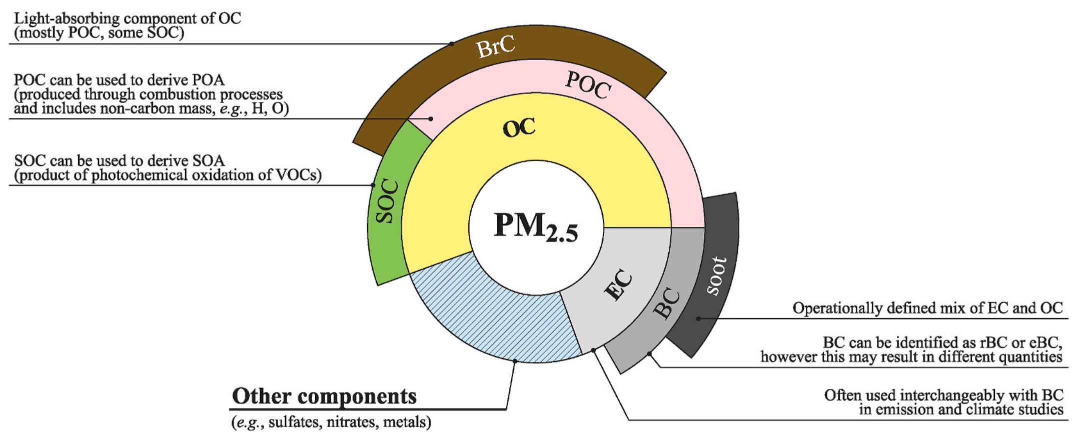

Fig. 1. Conceptual diagram of $PM_{2.5}$ composition summarizing Organic Carbon (Primary Organic Carbon, Secondary Organic Carbon, Brown Carbon), Elemental Carbon (Black Carbon reported as rBC/eBC depending on quantification method), and other constituents (sulfates, nitrates, metals).

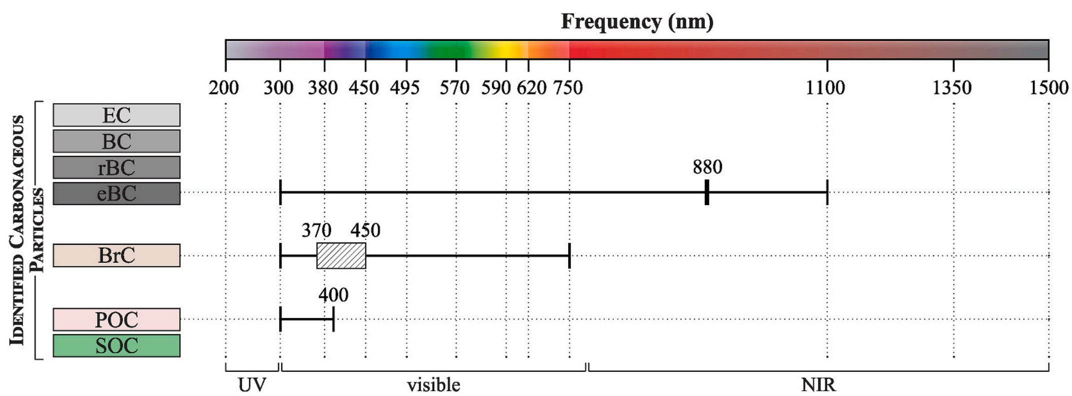

Fig. 2. Peak absorption of carbonaceous PM: Black Carbon (eBC/rBC) absorbs strongly across the visible and near infrared, peaking near 880 nm; Brown Carbon absorption increases toward shorter visible and UV wavelengths; Organic Carbon is mainly scattering, with absorption largely from Brown Carbon.

(Zhang et al., 2020). For BrC, Tao et al. (2024) report AAE values of 1.8–2.4, whereas Chen et al. (2020) report an annual AAE of 4.6 ± 0.8 for water-soluble BrC at 300–400 nm.

However, factors such as particle shape, size, and mixing state can affect AAE accuracy (Wang et al., 2020). The SSAAE helps differentiate aerosols based on their absorption properties (e.g., BrC tends to exhibit negative SSAAE values) (Chakrabarty et al., 2010; Zhang et al., 2020). Coating on BC particles induces a lensing effect that enhances light absorption. This phenomenon is quantified through E_{abs}. The coating effectively acts as an optical lens surrounding the BC core, concentrating incoming radiation and thereby increasing the overall absorption efficiency of the particle (Xu et al., 2018).

3. Comparison of methods for the characterization of carbonaceous particles

Accurate characterization of OC/EC is crucial for understanding their environmental and health impacts. Traditional techniques, such as TOA, provide detailed OC/EC ratios and insights into source apportionment but are expensive, time-consuming, and labor-intensive (Karanasiou et al., 2015; Shankar et al., 2022). Over the years, various optical techniques have been developed to address this need, each with distinct advantages and limitations (Manohar et al., 2021). This review examines both real-time monitoring and measurement methods (e.g., mass spectrometry (MS), in situ optical methods) and filter-based destructive (e.g., thermal-optical analysis) and non-destructive (e.g., light absorption-based methods, Fourier Transform Infrared Spectroscopy (FTIR), LII, and image-based techniques) methods for OC/EC characterization. Methods are classified into destructive or non-destructive depending on whether the sample remains intact for further analysis. The reviewed techniques are well-suited for AI integration, enabling automation, improved analysis, and the discovery of complex OC/EC patterns.

3.1. Real-time monitoring and measuring methods

3.1.1. Mass spectrometry

One of the real-time methods for measuring OC/EC in PM samples is aerosol MS (AMS), which involves the real-time detection and quantification of aerosol particle mass and chemical composition using a time-of-flight (TOF) mass analyzer (Ferge et al., 2006; Spencer and Prather, 2006; Zhou et al., 2016). The general principle is based on introducing aerosol particles into the instrument's inlet system, where desorption and ionization occur. The resulting ions are analyzed by a TOF MS, measuring their mass-to-charge ratio (Chen et al., 2024; Ferge et al., 2006; Healy et al., 2013; Salvador et al., 2016). Ionization is typically achieved through laser, thermal desorption, or electron ionization, as used in an aerosol TOF MS (ATOFMS), which produces a dual-ion mass spectrum for individual particles and reveals their chemical types and temporal variations (Healy et al., 2013; Spencer and Prather, 2006). Determining OC/EC using ATOFMS poses several challenges, stemming from the inherent complexity of aerosol composition, measurement techniques, and the difficulty of distinguishing between OC/EC (Ferge et al., 2006; Spencer and Prather, 2006; Zhou et al., 2016). Quantification difficulties arise from composition-dependent ionization efficiency and size-dependent particle detection efficiency, resulting in inconsistent sensitivities for particles of varying sizes or compositions (Healy et al., 2013; Zhou et al., 2016). Matrix effects further complicate the analysis, as the presence of other compounds within a particle can alter ion abundances, leading to inconsistent signals for the same chemical species. Additionally, variability in laser power introduces fluctuations in spectral peak intensities, which affect reproducibility (Healy et al., 2013; Zhou et al., 2016). Another limitation is incomplete sampling, which can potentially underestimate the total OC mass, while organic ion fragmentation may lead to EC overestimation (Laskin et al., 2012; Spencer and Prather, 2006). Differentiating OC and EC is further complicated by overlapping spectral peaks, low transmission efficiency, and selecting reliable marker ions (Ferge et al., 2006; Healy et al., 2013; Zhou et al., 2016). Furthermore, labile compounds' thermal decomposition leads to mass fragment misattribution, often requiring complementary techniques such as TOA for accurate quantification.

Despite these limitations, ATOFMS remains a powerful tool for investigating the composition and sources of carbonaceous particles, offering size-resolved chemical characterization (Casotto et al., 2022, 2023; Veld et al., 2023) and comprehensive chemical analysis by directly measuring individual particles' OC/EC ratio. Several studies indicate that ATOFMS can reliably produce results comparable to those obtained from conventional offline OC and EC measurement methods (Ferge et al., 2006). TOF-MS encompasses several techniques for the analysis of OC and EC, including High-Resolution (HR) TOF-MS (Casotto et al., 2022; Song et al., 2021, 2022; Twigg et al., 2023), Soot Particle (SP) AMS (Barreira et al., 2024), Proton-Transfer-Reaction (PTR) TOF-MS (Salvador et al., 2016), Extractive Electrospray Ionization (EESI) TOF-MS (Lopez-Hilfiker et al., 2019), and Single-Particle MS (SPMS) (Cho et al., 2021).

3.1.2. In situ optical techniques

In situ optical techniques offer real-time monitoring of BC concentrations with varying levels of sensitivity and specificity. The photoacoustic spectrometer enables direct, filter-free measurement of aerosol BC by detecting acoustic waves generated from the thermal expansion of air upon light absorption by particles. The amplitude of these waves correlates with BC concentration, providing high-sensitivity measurements; however, background noise remains a significant limitation, increasing the signal-to-noise ratio and detection limits (Schnaiter et al., 2023). Multi-angle absorption photometers (MAAPs) improve measurement accuracy by simultaneously detecting transmitted and scattered light at multiple angles, though their use is often constrained by the high cost and complexity of calibration (Massabò and Prati, 2021). Instruments such as particle soot absorption photometers (PSAPs) and multi-wavelength absorption BC instruments (MABIs) estimate BC concentrations using similar light attenuation principles, but face challenges related to scattering correction and non-linear attenuation under high filter loading conditions (Manohar et al., 2021; Massabò et al., 2013). Smoke stain reflectometers offer a low-cost alternative for BC assessment but lack the spectral resolution necessary for compositional differentiation (Yan et al., 2011). Multi-wavelength absorbance analyzers (MWAAs) allow for the discrimination of BC and chromophoric components, including BrC, through spectral absorption measurements, but their widespread application is limited by high costs (Massabò and Prati, 2021).

The Single-Particle Soot Photometer (SP2) is another in situ instrument designed to measure individual BC-containing particles' time-resolved scattering and incandescence signals as they pass through a continuous-wave laser beam. The SP2 employs distinct calibration standards to ensure measurement accuracy: polystyrene latex (PSL) spheres for the scattering channel and fullerene soot for the incandescence channel (Kong et al., 2024). Although this approach improves the quantification of rBC, uncertainties persist due to particle size detection limitations (J. Wang et al., 2022) and the lack of uniformity in calibration standards and mass loss correction procedures across studies (Zhang et al., 2023).

3.2. Destructive filter-based methods

Destructive filter analysis for OC/EC determination generally uses multi-stage sequential heating in a controlled environment, utilizing carbon analyzers with different TOA protocols, which rely on the thermo-optical transmittance (TOT) or reflectance (TOR) method (Bayramoğlu Karşı et al., 2020; Bhowmik et al., 2021; Demir et al., 2019; Górka et al., 2020; Luo et al., 2021; Samae et al., 2022; Samiksha et al., 2022; Xie et al., 2022). The separation of OC and EC fractions occurred at the "split point", where all OC fractions are removed from the filter, and the laser transmittance or reflectance signal returns to baseline (Bhowmik et al., 2021; Demir et al., 2019). The primary differences between the TOT and TOR methods are their operational principles and the resulting carbon concentration measurements, mainly due to distinct approaches to monitoring charring (Chow et al., 2004; Karanasiou et al., 2015; Yu et al., 2004). Karanasiou et al. (2015) summarized that many studies reported the TOT method yields EC concentrations 30–70% lower than those obtained by the TOR method, owing to differences in optical correction. The thermal protocol is another critical factor influencing the estimation of OC and EC concentrations in aerosol samples and should be tailored to the study tasks and particulate matter composition. The NIOSH, IMPROVE, and EUSAAR protocols are widely used for OC/EC measurements, showing comparable results for total carbon while differing in the OC/EC ratio (Karanasiou et al., 2015; Pan et al., 2022). Variations in results across protocols can be attributed to differences in optical correction for charring, the temperature programs employed (temperature, duration, and heating rate), and the chemical composition of PM samples (Karanasiou et al., 2015).

Several studies have noted that while OC measurements are generally comparable across protocols, EC concentrations can differ by 20%–48%, depending on the method (Merico et al., 2019; X. Zhang et al., 2022). Despite these differences, all three protocols demonstrate high repeatability, showing 5% uncertainty for OC and 10% for EC (Merico et al., 2019). However, the standardization of methods for OC and EC measurements remains a significant challenge for researchers, as differences in reported concentrations can impact source apportionment and atmospheric modeling.

While thermal-optical methods have enhanced accuracy, comparability, and the ability to analyze high mass loadings while ensuring precise differentiation between OC and EC, challenges remain. Merico et al. (2019) reported that seasonal measurement biases contribute to differences between protocols due to variations in meteorological conditions. Temperature calibration inaccuracies in thermal-optical analyzers also play a crucial role; discrepancies between actual and measured temperatures up to 50 °C can significantly impact the OC/EC split point selection (Conrad and Johnson, 2019; Pavlovic et al., 2014). Sampling artifacts add uncertainty to OC and EC measurements. Adsorption of gaseous organics on filters can overestimate OC concentrations by 10%–23%, while VOCs evaporation and premature gas evolution during analysis lead to underestimation and further distortion (Cheng et al., 2010). Discrepancies are also amplified by variations in calibration factors and methodologies across laboratories, leading to reported uncertainties of up to 26% for EC (Sipkens et al., 2024).

3.3. Non-destructive filter-based methods

3.3.1. Light absorption-based methods

Filter-based instruments (e.g., aethalometers, particle soot absorption photometers, and multi-angle absorption photometers) are widely used for measurements of optical properties of BC or EC in air pollution studies for their minimal complexity and "on-spot" optical analysis, making them suitable for continuous air quality monitoring (Zhao et al., 2020). Aethalometers (e.g., AE31, AE33 by The Magee Scientific Aethalometer®) are the most commonly used instruments with high sensitivity (e.g., <0.1 μg m^{-3}) and wide measurement range (e.g., 0–500 μg m^{-3}) (Sedlacek, 2016) that operate on the principle of light attenuation caused by carbonaceous particles collected on a filter (e.g., quartz filter fiber, Teflon-coated glass fiber) (Ferrero et al., 2021; Rajesh and Ramachandran, 2018). As aerosols are sampled, carbonaceous particles deposited on the filter absorb light at specific wavelengths (e.g., 370, 470, 520, 590, 660, 880, and 950 nm for the AE33), leading to a reduction in transmitted light intensity. The instrument measures mean BC concentration by analyzing light transmission through the sensing area (filter area with accumulated particles) and reference area (optical source stability checkpoint).

The aethalometer provides valuable high-resolution time data but is subjected to measurement bias due to inherent limitations such as multiple-scattering effects, filter loading, and absorption enhancement, requiring complex correction algorithms to account for matrix effects and light scattering by non-absorbing particles (Sedlacek, 2016; Wu et al., 2024). Zhao et al. (2020) indicated that utilizing unprocessed data, especially from earlier filter tape (Model 8050 in AE33), systematically overestimated b_{abs} and introduced instabilities related to the nonlinear filter-loading effect. One more limitation is that, despite aethalometers operate continuously, they do not provide instantaneous measurements. Instead, it requires the accumulation of PM on a filter over a specified interval before analysis. This process necessitates a waiting period for each collection cycle to obtain a sufficient sample load on the filter, making the measurement method indirect. Among single-wavelength techniques, a continuous soot monitoring system (COSMOS) is commonly applied for ambient BC monitoring since it applies an additional thermal method to remove VOCs and semi-VOCs fractions to quantify BC in the air accurately (Miyakawa et al., 2020).

3.3.2. Fourier Transform Infrared Spectroscopy (FTIR)

FTIR spectroscopy has been extensively used to characterize OC/EC in aerosol, offering a fast screening and non-destructive analysis of aerosol composition (Reff et al., 2007; Shankar et al., 2022; Yu et al., 2018), providing detailed information on functional groups and chemical bonding (Yu et al., 2018). In environmental studies, FTIR has been used to track seasonal variations in aerosol chemical properties (Górka et al., 2023; Shankar et al., 2022; Son et al., 2021; Yang et al., 2024). Winter samples exhibit stronger aromatic C–C and carbonyl bands, linked to biomass burning and oxidation, while summer spectra show more hydroxyl and carboxyl groups, indicating enhanced photochemical processing of organic matter (Kristensen et al., 2015; Voliotis et al., 2017). FTIR instrument equipped with attenuated total reflectance (ATR) allows direct measurements, minimizes light scattering, and does not require sample pre-treatment (e.g., KBr pellet), preventing sample loss and alteration (Takahama et al., 2016). Despite its many advantages, FTIR has several limitations that can lead to the inaccurate characterization of carbonaceous aerosols. One major challenge is spectral overlap, where absorption bands from different functional groups blend due to the complex aerosol composition, making it difficult to differentiate individual molecular species (Reggente et al., 2019). This is particularly important when analyzing condensed-phase samples, where broad peaks arise from similar bonds vibrating in slightly different chemical environments (Siciliano et al., 2018). Additionally, while FTIR provides valuable qualitative and semi-quantitative information, precise quantification of aerosol mass requires careful calibration with reference standards, which can introduce uncertainties in environmental applications (Russell, 2003; Rutherford et al., 2021). Moreover, using FTIR coupled with ATR requires careful baseline correction and precise sample handling to ensure accurate and reliable results, especially when working with filter-based samples (e.g., quartz filters) (Yu et al., 2018). The interpretation of spectra is also influenced by environmental factors such as humidity and sample matrix effects, which can alter the observed absorption features. Furthermore, FTIR cannot distinguish between different oxidation states of organic compounds, which limits its capacity to fully characterize SOA formation pathways.

3.3.3. Laser-induced incandescence (LII)

Laser-induced incandescence, widely employed to quantify rBC (Kong et al., 2024; Salo et al., 2024), is based on the short-term heating of individual light-absorbing particles to their incandescence (vaporization) temperature using laser radiation (Lee et al., 2025). A pulsed laser (at 1064 nm) rapidly raises the temperature of particles to 3000–4500 K, resulting in the emission of thermal (blackbody) radiation. The intensity and spectrum of this radiation depend on the particle's temperature, size, thermal conductivity, and composition. By analyzing the dynamics of changes in the intensity and spectrum of emitted radiation during cooling, it is possible to determine parameters such as particle concentration, size distribution, and morphology (Kong et al., 2024; Salo et al., 2024). The detection and quantification limits for light-absorbing BC particles using LII have been reported to be as low as 1 $\mu g\ m^{-3}$, demonstrating the method's high sensitivity for ambient air monitoring applications (Vasilatou et al., 2025).

3.3.4. Image-based techniques

The optical properties of carbonaceous species bound to aerosol particles allow for the use of image-based techniques to study their concentration. Typically, higher loadings of carbonaceous species increase the optical absorption at various wavelengths, depending on the species, including in the visible (380–780 nm) and ultraviolet (100–400 nm) ranges. Due to digital cameras' high availability and low cost, these tools can provide a significant advantage in studying carbonaceous aerosols. Ramanathan et al. (2011) first demonstrated a cellphone-based system for measuring BC, integrating a miniaturized aerosol filter sampler with a cell phone camera to measure the light reflectance in the red-light visible spectrum. The authors reported good agreement with a reference method, with a coefficient of determination (CoD) of 0.83, highlighting the feasibility of real-time BC monitoring. Du et al. (2011) adopted a similar setup but used a higher-resolution camera and measured the light transmittance through a filter, rather than reflectance. By comparing the intensity of the light of the loaded filter with a reference filter, the authors could estimate the concentration of BC with a CoD of 0.95–0.98. Another pioneering work (Cheng et al., 2011) introduced a scanner-based system for measuring elemental carbon, establishing a power-law relationship between Red-Green-Blue (RGB) values and EC loading, achieving a correlation coefficient of 0.85 (Forder, 2014). Also, a comprehensive comparison of optical techniques for the measurement of BC was conducted, including the use of scanners, which showed a strong correlation between the optical measurements and the reference method, with a CoD of 0.95 for EC loadings up to 100 $\mu g\ cm^{-2}$ (Forder, 2014).

Olson et al. (2016) applied color space analysis using digital images transformed into Hue Saturation Value (HSV) to separate luminance from color. The authors achieved a CoD of 0.92 for EC and 0.67 for OC quantification, demonstrating the potential of this technique for the measurement of other carbonaceous species, though with lower accuracy. However, the use of digital cameras for measuring carbonaceous aerosols has limitations. The accuracy of the measurements is highly dependent on the calibration of the camera and lighting conditions. Furthermore, digital cameras have built-in post-processing algorithms and compression techniques that can affect the accuracy of the measurements, outputting images that are not true representations of the aerosol samples. To address this, Chen et al. (2020) employed a standard color palette alongside the aerosol samples, maintained consistent lighting conditions, and applied raw image processing, resulting in a CoD of 0.904 for EC across nearly 2000 samples. Similarly, Jeronimo et al. (2020) incorporated a grayscale palette and a specially designed measurement chamber, reporting a strong correlation between image-based absorbance and the EC loadings (CoD of 0.99), which was higher than the Smoke Stain Reflectometer (SSR) reference method (0.98) (Chen et al., 2020; Jeronimo et al., 2020).

Image-based techniques have demonstrated significant potential for cost-effective and high-throughput monitoring of carbonaceous aerosols. However, the accuracy is highly dependent on the measurement setup and the calibration of the camera. Additionally, most studies have concentrated on measuring BC or EC, the most light-absorbing carbonaceous species. Future studies could focus on measuring other carbonaceous species, such as OC. Hyperspectral imaging across the UV–Vis–NIR range could further improve the differentiation and enable simultaneous quantification of multiple species.

The evolution of ML techniques, particularly convolutional neural networks, could enable the development of more complex models by learning complex patterns in the images, such as variations in environmental conditions or the presence of other aerosol species. Standardization of imaging protocols through controlled lighting conditions, the use of standard color palettes, and raw data acquisition could improve the accuracy of the measurements. Such consistency will also facilitate the development of specialized software for automated analysis and data interpretation.

4. Existing AI-based application techniques for the characterization of carbonaceous aerosols

4.1. AI applications for the filter-based thermo-optical method

A critical aspect of TOA analysis is correcting the OC's pyrolysis (charring) during the inert heating phase, which can lead to overestimating EC (Karanasiou et al., 2015). Although this is commonly addressed by continuously monitoring the optical properties of the filter, the accuracy of the OC/EC split can vary significantly depending on the various conditions. Despite this, direct ML applications aimed at improving the OC/EC split within the TOA method remain scarce. Existing studies primarily apply ML to post-analysis tasks, such as clustering and source apportionment using TOA data (e.g., 8C fractions) (Kumar et al., 2022), associating chromophores with specific fractions, or predicting TOA outputs from other measurements (Elcoroaristizabal and Amigo, 2021). This highlights the largely unexplored potential for ML to enhance the TOA method itself by using both thermogram and optical signals as input (Table 1).

4.2. AI applications for MS

Aerosol MS produces high-dimensional, time-resolved mass spectral data, which is well-suited for ML analysis due to its chemical complexity (Sandström et al., 2024). While traditional OA source apportionment methods, such as PMF, are widely used for AMS and Aerosol Chemical Speciation Monitor (ACSM) data (Vlachou et al., 2018), they rely on long-term, stable datasets and assume bilinearity and temporal invariance (Chen et al., 2018; Hristova et al., 2020). ML offers promising alternatives for more efficient, automated processing of AMS data in source apportionment applications.

Various supervised ML approaches have been applied to AMS data, often using PMF-derived factors. For instance, Pande et al. (2022) developed a two-step model that first classifies the probability of factor presence, then estimates fractional abundances through regression. This method does not require access to the full dataset and can be continuously updated as new data becomes available. Hybrid approaches combining traditional factorization with ML have also been explored. Simple regression and CMB models were used to reproduce PMF-derived OA factors, with CMB performing slightly better (Ng et al., 2011). Ensemble ML methods, including RF, Gradient Boosting, and SVR, improved prediction accuracy of OA factors by up to 30% in RMSE and 5% in CoD (Zhang et al., 2025), showing promise for near-real-time source apportionment. Additionally, PMF has been combined with unsupervised clustering (e.g., k-means) to classify OA subtypes without bilinearity assumptions (Äijälä et al., 2016).

SPMS methods, including ATOFMS, offer real-time, high-resolution chemical characterization of individual particles, making them well-suited for analyzing complex aerosol mixtures. The large data volumes

Table 1
AI-based approaches to overcome the limitations of optical methods in carbonaceous aerosol characterization.

Method	Existing AI-based applications	AI-based approaches to mitigate the limitations of optical methods	References
TOT/TOR	- BC source apportionment at urban sites. - AI clustering and source apportionment from TOA (8C), linking chromophores to fractions. - Predicting TOA outputs from other measurements	- Thermogram-based differentiation of OC/EC fractions - Automatic detection of OC-to-EC split point (RNN + transformers) - Calibration-curve optimization for robust OC/EC quantification across instruments and conditions - Data-driven standardization of temperature protocols from historical multi-lab datasets - With AI models trained on one dataset, one instrument can be adapted to other instruments and laboratories, improving cross-laboratory differences	Dillner and Takahama (2015); Elcoroaris-tizabal and Amigo (2021); Kumar et al. (2022); Pina et al. (2024)
MS	*SPMS* - SPMS single-particle spectra as multi-class classification (KNN, SVM, DT, RF, MLP) for aerosol typing - DL (CNN/RNN) for real-time spectral feature extraction and source attribution-Unsupervised clustering (k-means, DBSCAN) for particle grouping - Supervised ML (RF, SVM) for aerosol classification	- Novelty detection and real-time anomaly detection for unknown aerosol types - DRL for adaptive particle classification in dynamic conditions - Transfer learning for cross-laboratory SPMS dataset integration - Models trained on real-time air pollution data - AI-driven real-time anomaly detection in atmospheric particles - Predictive ML for tracking pollutant transport in air masses - Interpretable AI models for SPMS aerosol classification - AI models for learning aerosol patterns from unlabeled SPMS data	Beck et al. (2024); Boiko et al. (2022); Christopoulos et al. (2018); G. Wang et al. (2024); Xu et al. (2024)
	Bulk MS - Automated HRMS pipelines (MEDUSA): NNs for peak	- Self-supervised AI for compound discovery without spectral libraries	

(continued on next page)

Table 1 (*continued*)

Method	Existing AI-based applications	AI-based approaches to mitigate the limitations of optical methods	References
	detection, PCA for reduction, clustering/ classification - RF/GBTs for spectral pattern recognition and formula prediction - Deep learning (CNNs, Autoencoders) for peak picking and noise removal. - Rule-based and ML-assisted methods to enhance mass accuracy. - GNNs & Transformers for molecular formula assignment.	- Transfer learning for cross-platform MS spectra standardization - Active learning approaches for adaptive molecular classification - Self-supervised DL for MS/MS spectral generation - GNNs-based MS/ MS spectral prediction (MassFormer, 3DMolMS) - Transformer-Based Spectral Prediction (MassFormer, QC-GN2oMS2)	
Aethalometer	- AE/ meteorological data used to train both white-box (IAP, LASSO, ARIMA) and black-box (RF, SVM, CNN, LSTM) models as virtual sensors for BC prediction - BC monitoring network data used for model validation (NN and WRF) - Use of proxy models for gap-filling of BC time series - KRR and ANN improve prediction of BC optical properties across aging stages for fractal aggregates - Pretrained KRR/ ANN rapidly generate visible-range optical properties for specified BC fractal aggregates	- Improved radiative forcing estimates in climate models using ML-based BC optical property parameterizations across aging stages and fractal morphologies. - Development of a universal ML model to integrate all aerosols into climate models, combining core-shell and fractal-based aging representations. - Application of Bayesian ML methods (Gaussian processes) to predict BC optical properties with uncertainty estimation.	Fung et al. (2024); Gupta et al. (2024); Hamilton and Harley (2024); Ramshoo et al. (2024)
FTIR	- Fcg-Former used to resolve overlapping FTIR fingerprint peaks via self-attention mechanisms - CNNs used to classify organic functional groups from gas-	- Automated detection of OC/EC fractions based on FTIR peak patterns - AI-assisted tracking of seasonal changes in aerosol composition using DL models trained on FTIR datasets	Doan et al. (2024); (Enders et al., 2021)

Table 1 (*continued*)

Method	Existing AI-based applications	AI-based approaches to mitigate the limitations of optical methods	References
	phase FTIR using image-based spectra	- Real-Time Spectral Classification for OC/EC Aerosols - Hybrid CNN + Transformer resolves overlapping OC/EC peaks and improves separation of SOA vs POA - Autoencoders + contrastive learning reduces need for labeled-data and adapts to new spectral datasets - Denoising autoencoders + GANs enhance weak signals and enable detection of trace OC components	
LII	- Prediction of rBC concentrations, emissions or particle sizes based on raw LII signals in real time - More realistic estimation of uncertainties leads to increased confidence in LII data - Eliminating background noise from LII signals	- Optimization of calibration using the experimental data and adjustment based on changes in laser fluence and detector responses - AI can be applied to classify LII signals based on the different rBC sources of emissions - Possible integration of AI in the diagnostic of combustion using LII data of rBC concentrations	Hadwin et al. (2018)
Image-based techniques	- Linear interaction models - Color space transformations - Power-law relationships	- CNN used to learn complex patterns beyond linear relationships - Tree-ensemble models for carbonaceous component estimation in data-scarce scenarios - Unsupervised autoencoder features from filter images without traditional loading measurements. - Image-based source classification (traffic, biomass burning, industrial) without complex chemical analysis	(Chen et al., 2020; Jeronimo et al., 2020; Olson et al., 2016; Ramanathan et al., 2011)

generated are also ideal for ML applications (Table 1). Unsupervised clustering methods, particularly Adaptive Resonance Theory 2 (ART-2A), have become standard for SPMS analysis (Phares et al., 2001; Y. Wang et al., 2024; Xu et al., 2024), with k-means and its variants also explored (Bi et al., 2014; Freney et al., 2006). These methods group particles with similar mass spectra, using cosine similarity in ART-2A and Euclidean distance in k-means. ART-2A also includes a "vigilance" parameter that controls how similar particles must be to form a new cluster.

Density-based methods such as DBSCAN (Density-Based Spatial Clustering of Applications with Noise) may offer advantages for identifying non-convex clusters in SPMS data (Zhou et al., 2006). While unsupervised clustering methods have been widely applied to SPMS data, a significant limitation is the need for substantial manual post-processing. Interpreting cluster outputs, assigning chemical meaning, and linking them to emission sources depends heavily on expert judgment. Moreover, results can be sensitive to algorithm settings, such as the vigilance parameter in ART-2A or the predefined number of clusters in k-means, which adds further subjectivity to the analysis (Zhao et al., 2008).

To overcome the limitations of unsupervised clustering, considerable effort has been made to automate particle classification. Studies have employed models such as K-nearest neighbors (KNN), Support Vector Machines (SVM), and Feedforward Neural Networks (FNN), among others (Table 1) (Xu et al., 2024). These models are trained on labeled SPMS data (based on laboratory standards, source samples, or expert-labeled). Supervised approaches have achieved high classification accuracy, exceeding 90% in some cases (Christopoulos et al., 2018; Xu et al., 2024), particularly when categorizing particles into broader groups (e.g., soil, mineral, biological), with lower performance for finer subtypes (Zhao et al., 2008). A common practice involves applying a probability threshold to assign uncertain particles to an "unclassified" category for further review.

Despite their strong performance, supervised ML methods rely heavily on large, high-quality labeled datasets, which are often difficult to obtain for SPMS applications. Moreover, the dynamic and evolving nature of aerosol chemical composition can limit model generalizability and hinder the detection of novel or emerging sources. These limitations underscore the need for complementary approaches that enable high-throughput classification and automatic flagging of anomalous or previously unrecognized particle types. Adaptive learning strategies may be critical in enhancing model flexibility and responsiveness to real-time data.

4.3. AI applications for light absorption-based methods

Light-absorption instruments are subject to several limitations. These include uncertainties in data interpretation due to the lack of standardized measurement protocols (Petzold et al., 2005), instrumental limitations related to software design and correction algorithms (e.g., for light scattering and filter loading effects) (Luoma et al., 2021), potential data loss resulting from instrument malfunctions (Zaidan et al., 2019), and challenges in capturing reliable temporal trends due to variability in atmospheric conditions and emission sources (Collaud Coen et al., 2020). Several attempts have been made in recent literature to address these issues with AI-based applications. Fung et al. (2024) trained both white-box (e.g., IAP, LASSO, ARIMA) and black-box (e.g., RF, SVM, CNN, LSTM) models on aethalometer and meteorological data to serve as virtual sensors for BC prediction (Table 1). Black-box models showed better performance and higher prediction accuracy, with LSTM yielding the best results. Hamilton and Harley (2024) developed a model that combined high-resolution BC monitoring network data and source-resolved BC concentrations generated by the Weather Research and Forecasting (WRF) model.

A neural network was trained on WRF-derived outputs and accurately predicted BC concentrations and source contributions for 100 summer days. The model achieved results comparable to the full WRF model, reducing computational cost by approximately 80% and offering an efficient alternative to large-scale monitoring. Gupta et al. (2024) evaluated the performance of several ML models, including RF, support vector regression (SVR), multilayer perceptron (MLP), and ANN, in predicting BC mass concentrations. Among the evaluated models, the MLP demonstrated the highest performance across seasons and locations, with strong predictive accuracy, highlighting its potential for gap-filling in BC time series and cost-effective estimation of BC concentrations. Romshoo et al. (2024) developed an ML-based framework to predict the optical properties of BC fractal aggregates at different atmospheric aging stages using input features related to particle size, morphology, and coating. Kernel Ridge Regression (KRR) and ANNs accurately predicted key optical parameters, including SSA (ω) and MAC, outperforming conventional Mie theory-based approaches.

4.4. AI applications for FTIR

Vibrational spectroscopy data (e.g., FTIR and Raman spectra) is also well-suited for applying AI and ML techniques, enabling effective data classification and identification of key absorption regions. FTIR spectral data were utilized in near-infrared hyperspectral imaging (HSI-NIR) and analyzed using AI-based multivariate techniques, including Principal Component Analysis (PCA) and Multivariate Curve Resolution (MCR), to isolate spectral contributions from both filter material and aerosol particles. Additionally, a Partial Least Squares (PLS) regression model was employed to predict OC, EC, and total carbon (TC) concentrations from the spectral data, demonstrating the potential of AI-assisted spectral interpretation in carbonaceous aerosol analysis (Elcoroaristizabal and Amigo, 2021). Doan et al. (2024) introduced the Fcg-Former model, which integrates a novel self-attention mechanism, designed to capture long-range dependencies within FTIR spectral data (Table 1). This advancement addresses a critical limitation in traditional spectral analysis, specifically, the overlapping absorption peaks in the fingerprint region (400-1500 cm^{-1}), which often hinders accurate interpretation of functional groups. The model effectively identifies subtle spectral features that are otherwise difficult to distinguish due to the spectral complexity. As a result, the Fcg-Former can reliably identify 17 distinct functional groups from FTIR spectra, significantly improving spectral deconvolution and functional group classification over conventional methods.

Authors (Enders et al., 2021) developed a generalized model via CNNs to identify 15 of the most common organic functional groups in gas-phase FTIR spectra by treating the spectral data as image-based inputs rather than relying on raw numerical formats. This approach uses CNNs to classify functional groups by recognizing patterns in 2D spectral images. Framed as a multi-label classification task, the model predicts the presence or absence of each group by learning respective absorption features from transformed intensity-frequency data, reducing the need for manual interpretation. Riad et al. (2019) evaluated multiple ML algorithms for VOC detection using high-resolution FTIR spectra and found that SVM and ensemble methods outperformed simpler classifiers (e.g., ensemble classifiers). VOCs identification accuracy exceeded 90% with high signal-to-noise (SNR) data, even with small training sets; however, it declined for gas mixtures due to overlapping absorption features.

5. Future perspectives and recommendations

Despite notable AI-based OC/EC analysis progress, machine and deep learning approaches face key limitations. Data scarcity remains a significant challenge due to the cost and effort of acquiring labeled datasets. Additionally, the limited interpretability of deep models hinders result transparency for domain experts, while high computational demands restrict their use in field applications. Emerging solutions include semi-supervised and active learning to reduce data annotation needs, explainable AI to enhance model interpretability, and TinyML for deploying lightweight models on low-resource devices. These strategies offer promising directions for more data-efficient and scalable OC/EC analysis. However, majority of studies remain proof-of-concept demonstrations with small, non-standardized datasets, raising concerns about model generalizability and reproducibility across instruments, regions, and atmospheric conditions. This section discusses these approaches and their potential for advancing future research in data-efficient and resource-aware AI applications for OC/EC analysis. Fig. 3 presents a roadmap for improving OC and EC identification by implementing AI-driven tools, ultimately enhancing the accuracy of air pollution assessment.

5.1. Semi-supervised learning and active learning (AL)

The risk of propagating labeling biases from TOA to ML frameworks remains largely unaddressed, and data classification remains one of the major challenges in aerosol science. The effectiveness of the classification task heavily depends on access to an "oracle", a source of ground-truth labels, such as a human annotator or a cost-driven system, that is available during training (Settles, 2009). When such access is restricted to an initial labeled subset, semi-supervised learning (SSL) becomes essential, specifically designed to use structures in both labeled dataset (L) and unlabeled data (U) to build models that outperform those trained on L alone, which is needed when conventional labeling (e.g., via TOA) is not feasible at scale (van Engelen and Hoos, 2020).

In contrast, AL assumes continuous oracle access and selective labeling of the most informative data points in U. By focusing on instances near decision boundaries, AL achieves higher accuracy with fewer labeled samples, since not all unlabeled instances equally enhance learning (Aggarwal, 2014). The practical integration of SSL and AL with low-cost sensors such as aethalometers, is promising but has yet to demonstrate quantitative equivalence with standardized OC/EC reference methods as translating measurements from these instruments into actionable OC/EC estimates still relies on TOA-based ground truth. To address this, TOA can be selectively applied to a small subset of aethalometer data to create a labeled set, while the remaining unlabeled data can be used in SSL or AL frameworks to develop accurate models that map aethalometer outputs to reliable OC/EC estimates.

5.2. Incremental learning (IL)

In low-data environments or real-world monitoring settings, labeled or unlabeled data is often acquired incrementally over time rather than all at once (Polikar et al., 2001), posing a challenge to conventional static learning approaches that assume access to complete datasets for model training. Incremental learning has been proposed as a solution for a continuous data enabling learning from new data while preserving prior knowledge (Shaheen et al., 2022; L. Wang et al., 2024). IL is defined by three key features: (i) updating the model with each new data instance, (ii) retaining previously learned knowledge, and (iii) eliminating the need to access earlier training data (Polikar et al., 2001).

Although, IL approaches showed conceptual value for continuous monitoring, its practical application in environmental contexts, especially in OC/EC analysis, remains largely unexplored. IL currently face the unresolved problem of catastrophic forgetting that can occur when new data significantly diverges from previously observed data, which distorts long-term trend analysis. To maintain robust model performance, a balance must be achieved between plasticity (i.e., the ability to acquire new information) and stability (i.e., the preservation of previously learned knowledge) (Luo et al., 2020). In OC/EC analysis IL can support continuous model updates using data from instruments such as aethalometers or SPMS, although there is also insufficient investigation of how environmental variability (e.g., humidity, temperature, and seasonal source shifts) affects the model's stability-plasticity balance.

However, there are examples of successful implementation of IL to traditional PMF methods, so-called rolling PMF. This method effectively captures aerosol source dynamics in multi-season data and overcomes PMF limitations (Chen et al., 2022). Additionally, IL could be useful for correcting instrumental drift and calibration, allowing the model to learn from new data to adjust the calibration parameters and ensure that

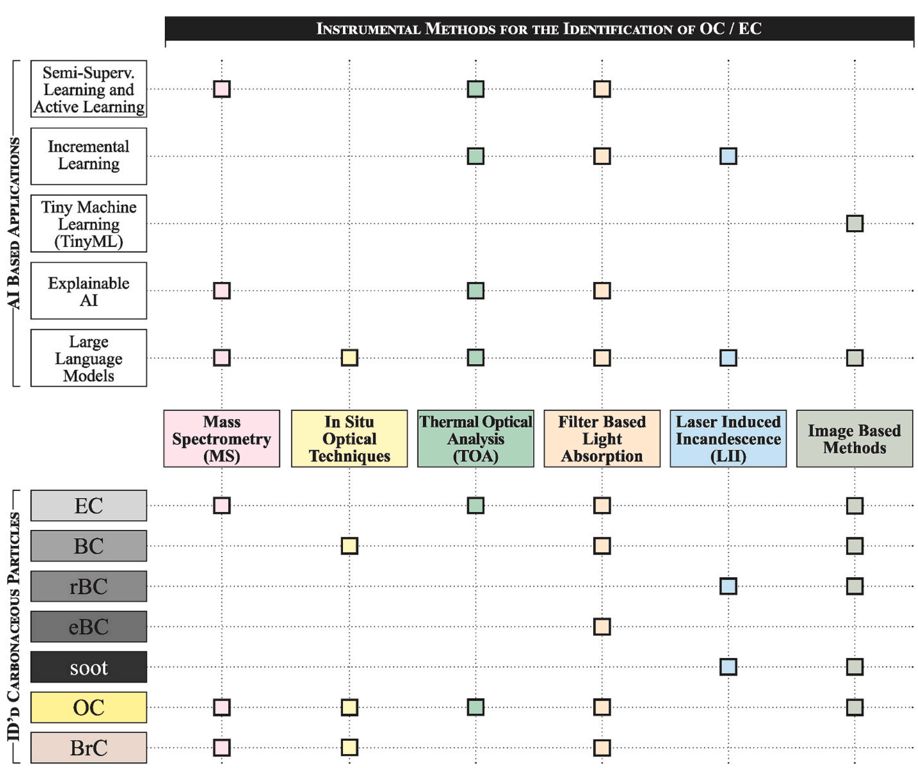

Fig. 3. Schematic diagram linking AI approaches (top), conventional instruments (center), and carbonaceous particle types (bottom), showing how methods can target specific organic fractions and how AI can enhance detection, quantification, and interpretation.

incoming data is represented accurately. This can be particularly important in applications where instruments are susceptible to changes in environmental conditions, which is often the case for methods such as aethalometers.

5.3. Tiny machine learning (TinyML)

Advances in embedded sensor technologies have enabled real-time applications of ML, giving rise to a new subfield known as TinyML that integrates AI with hardware-aware optimization (Capogrosso et al., 2024; Dutta and Bharali, 2021; Lin et al., 2023). The primary goal of TinyML is to bring the capabilities of AI to edge devices for real-time applications with minimal energy consumption and to focus explicitly on deploying lightweight and efficient learning models on edge devices, which operate with only kilobytes to a few megabytes of memory and under strict power and energy constraints.

For example, applying TinyML in image-based OC/EC analysis enables on-site aerosol sample examination, providing immediate feedback on chemical composition (Chen et al., 2020; Jeronimo et al., 2020). Even though image-based OC/EC analysis already enables visual estimation of carbon loading from optical images of filters, deploying this capability in the field has historically been limited by computational requirements and energy consumption (Jeronimo et al., 2020; Ramanathan et al., 2011). TinyML overcomes these limitations by enabling ML inference on microcontrollers consuming milliwatt-range power, which allows battery-powered or energy-harvesting devices to perform on-premises analytics (Schizas et al., 2022). TinyML becomes particularly advantageous when transmitting data to high-performance computing systems is impractical or undesirable (Baciu et al., 2025; Trigkas et al., 2025), such as in environmental sensing (Andrade et al., 2022) and industrial monitoring (Antonini et al., 2023). For OC/EC detection, TinyML enables data processing directly on edge devices instead of sending large volumes of raw data (e.g., mass spectra) to a central system for analysis, allowing for the local inference of source types, concentration trends, or anomalies in real time.

TinyML is crucial for portable devices in remote areas where data transmission to a central server is impractical. The balance between power efficiency and model accuracy has not been well examined, and the lack of common benchmarks for on-device evaluation makes it difficult to compare results across studies. Furthermore, integrating TinyML with optical or imaging-based OC/EC platforms requires interdisciplinary expertise in embedded systems, which is not always the case for atmospheric research groups, slowing translational progress. Integrating TinyML with automated sampling can enable continuous real-time aerosol monitoring and potential on-device model updates for changing conditions (Elhanashi et al., 2024).

5.4. Explainable AI (XAI)

XAI (Ali et al., 2023; Bommer et al., 2024) provides a crucial alternative to the “black-box” nature of deep neural networks, revealing the rationale behind a model’s decisions and increasing its transparency. It remains largely unexplored in OC/EC research, even though the interpretability is critical in environmental studies, where understanding model behavior is essential (Antonini et al., 2023). However, several successful applications of XAI have also been reported.

For example, XAI methods have been used to understand the aging process of atmospheric BrC, offering insights into the individual chromophores contributing to BrC absorption (Laskin et al., 2025; Y. Wang et al., 2024). Interpretable ML models using the SHAP (SHapley Additive exPlanations) method have successfully correlated 38 typical chromophores with BrC optical properties. Given that BrC is a complex mixture of light-absorbing organic compounds, XAI approaches enable interpretation of ML models that connect molecular composition data with optical behaviors, revealing key chromophore classes such as nitroaromatics, imidazoles, polyaromatic hydrocarbons, and quinones (Y. Wang et al., 2024). These methods help to elucidate how aging modifies BrC chromophores, improving understanding of their climate and air quality impacts (Laskin et al., 2025). Moreover, by using XAI methods, researchers can uncover previously overlooked signals, features, and patterns in data, leading to new insights and scientific hypotheses.

Techniques such as SHAP and LIME (Local Interpretable Model-agnostic Explanations) can explain individual predictions, aiding in the understanding of the factors that lead to specific classifications or predictions. Although these techniques provide detailed explanations, they can also be computationally intensive and may yield inconsistent results when applied to high-dimensional data (Kalasampath et al., 2025).

5.5. Large language models (LLMs)

LLMs and multimodal foundation models, such as GPT-4 (Open AI, 2023), Google’s Gemini (Gemini TeamGoogle, 2023), and Meta’s LLaMA (Touvron et al., 2023) introduce unprecedented flexibility for integrating heterogeneous environmental data. While LLMs have been commonly applied in natural language processing, recent studies have shown the capability of LLMs for analyzing and forecasting time series, detecting patterns, structures, and semantics in the input data. For example, time-series data are divided into patches, which are then embedded and mapped into a language-like representation using a small set of learned text prototypes (Jin et al., 2023). Jin et al. (2023) showed that LLMs can identify emerging research opportunities in environmental science, while task-specific prompts can supply context and domain knowledge, enabling the model to reason over time-series data similarly to NLP tasks. Additionally, LLMs can be used for OC/EC estimation, and their multimodal capability to process text, images, and tabular data simultaneously makes them transformative tools in carbonaceous aerosol analysis. LLMs can integrate diverse data types (e.g., aethalometer outputs and time-series meteorological data) and image-based measurements to derive meaningful insights into pollution dynamics (H. Li et al., 2024). Simultaneously, multimodal LLMs can help contextualize and interpret the data more accurately than single-modal analysis alone. For example, models such as EarthGPT have successfully unified multiple remote sensing tasks and sensor modalities for land surface interpretation (Zhang et al., 2024). Similar multimodal architectures could potentially be adapted to the characterization of carbonaceous aerosol by combining filter images, spectral data, mass spectrometry results, and thermal-optical analysis outputs. Such integrated approaches could enable comprehensive characterization of OC, EC, BC, and BrC by fusing information from diverse analytical techniques, potentially improving understanding of aerosol sources, morphology, chemical composition, and optical properties crucial for evaluating their roles in climate forcing and air quality impacts.

Lastly, LLMs can serve as auxiliary tools for report generation and interpretation. For example, Song et al. (2025) showed that LLMs can effectively generate precise regulatory information, perform data analysis, and provide location-specific recommendations relevant to atmospheric studies. As such, complex data outputs such as spectral plots can be converted into descriptive summaries or alerts suitable for communications with experts or the public (de Curtò et al., 2024). This capacity aligns with the increasing demand for real-time interpretable outputs from air quality monitoring systems. However, LLMs still require human input when handling complex parameter optimization. In this context, combining human expertise with LLM-based frameworks offers a more effective approach to managing data-intensive tasks (Nie and Liu, 2025).

6. Conclusion

This review highlights that traditional methods, including TOA, MS, light absorption instruments, and FTIR, remain fundamental for analyzing carbonaceous particles in PM. However, the accurate characterization of OC and EC still faces challenges such as labor-intensive procedures, high operational costs, measurement uncertainties, and complexities in data interpretation. Recent studies indicate that AI can effectively address these limitations by improving spectral deconvolution, automating data processing, and enhancing source apportionment and predictive accuracy. Advances in SSL, AL, IL, and TinyML provide practical solutions to overcome data scarcity, adapt to dynamic environmental conditions, and enable real-time, energy-efficient AI on resource-constrained edge devices. Furthermore, the integration of XAI and LLMs introduces new opportunities for interpretability, automation, and multimodal reasoning, fostering more transparent and comprehensive environmental monitoring. Despite these advancements, further research is needed to standardize methodologies, expand reliable training datasets, improve model generalizability across instruments and environments, and establish robust validation frameworks. The findings highlight the strong potential of integrating AI-driven frameworks with conventional analytical techniques for characterizing carbonaceous aerosols, providing an efficient solution to enhance urban air quality assessments. This method also aids broader initiatives toward sustainable urban development by facilitating more effective air quality management strategies and decision-making.

CRediT authorship contribution statement

Akmaral Agibayeva: Writing – original draft, Visualization, Investigation, Data curation, Conceptualization. **Aset Muratuly:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Olga P. Ibragimova:** Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Kazbek Tursun:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Azamat Mukhamediya:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Amin Zollanvari:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Data curation, Conceptualization. **Akhan Almagambetov:** Writing – review & editing, Visualization. **Serpil Yenisoy:** Writing – review & editing. **Nassiba Baimatova:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Data curation, Conceptualization. **Ferhat Karaca:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available upon request.

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